1. Clinical Question & Core Hypothesis

Clinical Question

Radiomic markers of metastatic lesion cellularity/heterogeneity (e.g., low ADC25 and high texture entropy) variably predict survival in mCRPC; why do some patients with imaging-suggestive aggressive lesions nonetheless have better outcomes, while others fare poorly? 2) Tier-1 rationale used: (a) Imaging ↔ proliferation link — diffusion MRI low-ADC and high texture heterogeneity reflect high cellularity and necrotic/heterogeneous tumor microenvironments associated with aggressive biology; (b) AR signaling vs proliferation — AR activity can mark tumors that remain androgen-driven and therapy-sensitive, whereas high proliferation (MKI67) signals aggressive, potentially AR-independent clones. I combined these two core paths to ask what modulates the prognostic impact of imaging. 3) Tier-2 supporting evidence: DNA-repair/genomic instability (BRCA2, ATM) may enable rapid clonal evolution; neuroendocrine markers (CHGA, SYP) reflect lineage plasticity that decouples AR signaling from imaging phenotype; immune/inflammation signatures can modify microenvironmental effects on diffusion and heterogeneity. 4) Endpoint choice: Overall survival (OS) was selected as the single primary endpoint because it is the most definitive, least-biased outcome in mCRPC and aligns with the clinical goal of risk stratification for therapy sequencing. 5) Deliberate exclusions: I excluded the full high-dimensional radiomics catalogue (e.g., all GLCM/run-length metrics, fractal dimension, PSMA-PET SUV metrics) from the core hypothesis to keep focus on a single, mechanistically interpretable imaging feature (ADC25). I also excluded neuroendocrine and DNA-repair markers from the core claim (they are secondary modifiers) to preserve the 1-imaging + ≤2-anchor constraint?

Working Hypothesis

In metastatic castration-resistant prostate cancer (mCRPC) bone metastases, the prognostic value of low ADC25 diffusion imaging is dependent on tumor proliferative and AR signaling status. Specifically, lesions with low ADC25 and high MKI67 expression together with preserved AR signaling identify a high-risk subpopulation with significantly shorter overall survival; when MKI67 is low or AR signaling is disrupted (e.g., AR-V7–positive status), the association between ADC25 and OS is attenuated, indicating AR-independent or less proliferative biology. Secondary modifiers—BRCA2 and ATM alterations, CHGA/SYP (neuroendocrine lineage), AR-V7 status, and immune signatures—modulate the strength and direction of this imaging–gene interaction and can refine risk stratification beyond the two-anchor core.

This hypothesis posits that combining a single, mechanistically interpretable imaging feature (ADC25) with two core molecular anchors (MKI67 and AR activity) will yield a more precise prognostic model in mCRPC and reveal biologically distinct subclones that drive differential survival outcomes.

Element	Value
Imaging feature	ADC 25th percentile (low ADC within metastatic lesions, mpMRI diffusion metric)
Anchor gene(s)	AR (AR signaling activity), MKI67
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR (AR signaling activity): Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

MKI67: MKI67) signals aggressive, potentially AR-independent clones.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	BRCA2	Supporting modifier from knowledge graph
Secondary	ATM	Supporting modifier from knowledge graph
Secondary	CHGA	Supporting modifier from knowledge graph
Secondary	SYP	Supporting modifier from knowledge graph
Secondary	AR-V7 status	Supporting modifier from knowledge graph
Secondary	inflammation/immune signature	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- BRCA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ATM: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.